

The SharpCorePDF Manual

VERSION 1.0.0 · .NET 10 AND .NET 11

Merge PDFs in .NET without a third-party engine: page selection, bookmarks, metadata, encryption and inspection — behind a domain you can reason about.

- → The whole public API, from MergePlan to MergeResult
- → Page selections, output policies, encryption and document inspection
- → Errors, warnings and dependency injection in practice
- → A cookbook of recipes for services that merge documents for a living

SharpCorePDF is published as the SharpCorePDF meta-package, with separate SharpCorePDF.Domain, .Application, .Infrastructure and .DependencyInjection packages on nuget.org.
This manual describes version 1.0.0 exactly.

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Preface

SharpCorePDF merges PDF documents in .NET. It reads the documents you give it, copies the pages you select into a new document, and writes that document without calling a native library, a command line tool or a third-party PDF engine. It also inspects documents, so a service can learn what it is about to merge before it commits to merging it.

This manual describes version **1.0.0** of the library and the public API of that version. It is written for the developer who has to ship a feature, not only for the developer who wants to read source code: every chapter explains the behaviour you can rely on, the behaviour you should not rely on, and the code that exercises it.

Who this manual is for

You will get the most out of it if you are comfortable with C#, asynchronous methods and dependency injection. No knowledge of the PDF specification is assumed. Where a term from the specification matters — outline, page label, document information, security handler — the chapter explains it in the terms a caller needs, and stops there.

How the manual is organised

The first four chapters are the shortest path to a working merge:

- **Chapter 1** describes what the library does and the guarantees it makes.
- **Chapter 2** states, just as plainly, what it does not do.
- **Chapter 3** installs the package and merges three documents.
- **Chapter 4** explains the merge plan, the object at the centre of the API.

Chapters 5 to 10 cover one part of the API each: page selection, output options, encryption, inspection, errors and warnings, and dependency injection. Chapters 11 and 12 are for readers who want to understand the design or who need a recipe they can adapt: the architecture, then a cookbook.

You do not have to read the chapters in order. Each one starts from a complete example and refers to earlier chapters only for background.

Conventions

Page numbers are one-based. Page 1 is the first page of a document, and every page number that crosses the API — in a page selection, in `PdfPageInfo.Number`, in an error message — is one-based. The PDF format itself numbers objects from zero internally; that detail never reaches you.

Code is complete enough to compile. Snippets omit `using` directives and surrounding class declarations when they are obvious, but every method call, property and argument name is the one in the 1.0.0 API. The cookbook contains full methods for the recipes that need more than a fragment.

Warnings are not errors. A document that was repaired while being read still merges. A chapter that had to drop a bookmark still produces a result. Whenever the library makes such a decision it records a `PdfWarning`, and Chapter 9 explains which warnings mean what. Treat the `Kind` of a warning as the contract, never the wording of its message.

The manual is built from Markdown. The chapters in `docs/manual/src` are the source of both the GitHub documentation and the PDF. If you find a discrepancy between this text and the code, the code is the source of truth; please open an issue so the text can be corrected.

> **Version note.** This manual is written against 1.0.0. The public API is covered by an approved snapshot in each project, so a change to a public member is a deliberate, reviewed event rather than an accident of a refactoring. Code that compiles against 1.0.0 keeps compiling across a patch release, and the changelog records every addition and every change of behaviour.

What you need

- The .NET SDK that can target **.NET 10** or **.NET 11**. The packages ship assemblies for both.
- A PDF to merge. Any reader can produce one; the tests use documents written by a different producer precisely so that the library is not tested only against its own output.
- For building this manual: Python with `markdown`, `pygments` and `pypdf`, and Chrome or Edge. That is a maintainer concern, not a requirement of the library.

1. What SharpCorePDF Is

SharpCorePDF is a PDF **merging** and **inspection** library for .NET. It answers two questions:

1. *What is in this document?* — page count, page sizes, rotation, metadata, outlines, annotations, form fields, embedded files, encryption and the warnings the reader had to record.
2. *How do I produce one document from several?* — in the order I choose, with the pages I choose, under the output policies I choose.

Everything else in the API exists to make those two operations predictable: an immutable plan, value objects for the inputs, read models for the outputs, typed exceptions for the failures, and warnings for the decisions that are not failures.

Two use cases, one engine

Merging and inspection share the same reader. A merge inspects every source anyway — it has to know the page count before it can resolve `"2-5"` — so inspection is not a separate code path with its own interpretation of a damaged file. If `InspectAsync` reports four pages, the merge that follows sees the same four pages and the same warnings.

C#

```
using SharpCorePDF.Application.Extensions;
using SharpCorePDF.Domain.Aggregates;
using SharpCorePDF.Infrastructure.Extensions;

var plan = MergePlan.Create()
    .AddFile("quarterly-cover.pdf")
    .AddFile("quarterly-body.pdf", pages: "1-12")
    .AddFile("quarterly-appendix.pdf", displayName: "Appendix");

var result = await merger.MergeToFileAsync(plan, "quarterly-report.pdf");
```

The call above produces a new file and returns a `MergeResult`:

TEXT

```
18 pages from 3 sources, written as PDF 1.7, 421,338 bytes, in 41 ms
Primary source: quarterly-cover.pdf
Warnings: none
```

The result is a *summary*, not the document. It tells you how many pages were written, how long it took, which source led the merge, what each source contributed and which warnings the operation produced. That is enough for a log line, an audit record or a UI that says "18 pages, 3 documents".

What a merge actually does

A merge is not a concatenation of files. The engine:

- reads each source, following the page tree to its leaves and resolving indirect objects;
- copies every object the selected pages need into a new document, rewriting references so that the copies refer to each other and not to the source file;
- builds a new page tree in source order, preserving each page's size, rotation, content, resources and annotations;
- merges bookmarks into one outline whose top-level entries follow source order;
- applies the output policies: document information, page labels, form fields, embedded files, encryption and the PDF version written in the header;
- writes a complete file with a cross-reference table and a trailer.

The source files are only ever read. The merge does not modify them, does not need write access to them, and does not leave temporary files beside them.

The guarantees you can build on

The plan is immutable. Every method that changes a plan returns a new plan; nothing mutates an existing one. A plan can be built at start-up, validated once, shared between requests and executed concurrently without a lock. Two plans with the same sources and options are equal.

Plans are validated before work starts. `Validate()` reports every problem it can see without opening a document — no sources, a duplicate identity, an unsupported target version, an encryption algorithm that the target version cannot carry. `EnsureValid()` turns those problems into a single `MergePlanValidationException`. A bad plan fails before any output file is created.

Failures are typed. A malformed source, a missing password, an unsupported feature and an invalid plan each have their own exception type, all derived from `PdfException`. You can catch the base type for a coarse handler or the derived types for a precise one.

Recoverable problems are warnings. The reader rebuilds a missing cross-reference table, skips an unreadable bookmark and assumes a page size when the page tree does not state one. Each decision is recorded as a `PdfWarning` with a stable `Kind`, so a caller can decide whether the document is good enough without parsing error text.

Cancellation is honoured. Every asynchronous operation takes a `CancellationToken`. Cancellation throws `OperationCanceledException`, the .NET convention, and does not masquerade as a PDF error.

No native dependency. The package is managed code. There is no `pdfium`, no `qpdf`, no executable to deploy, no platform-specific runtime to select and nothing to install on the host beyond .NET.

The packages

The installable package is **`SharpCorePDF`**. It is a meta-package: it carries no assemblies of its own and depends on the four layers, so one `dotnet add package SharpCorePDF` is the whole install.

Package	Contains	Reference it when
`SharpCorePDF.Domain`	Plan, sources, value objects, read models, ports, exceptions, warnings	You build plans in a domain project that must not depend on the engine
`SharpCorePDF.Application`	Merge and inspection use cases, convenience extensions	You need the services without the concrete engine
`SharpCorePDF.Infrastructure`	Reader, engine, file and memory sources, engine options	You construct the engine yourself instead of using DI
`SharpCorePDF.DependencyInjection`	`AddSharpCorePDF()` and configuration binding	You register the library in a `IServiceCollection`
`SharpCorePDF`	Dependencies on all four	You just want the library

The layers exist so that a solution can keep its domain free of infrastructure, not so that every consumer has to think about them. If you are unsure, install `SharpCorePDF`.

What "no third-party engine" means in practice

The PDF parser, the object copier, the outline merger, the encryption writer and the cross-reference writer are implemented in this repository. The runtime dependencies of the library projects are the `Microsoft.Extensions.*` abstractions used by the dependency injection package; there is no PDF library underneath.

That has consequences worth knowing before you adopt it:

- The feature set is deliberately bounded. Chapter 2 lists the boundary.

- The writer emits classic cross-reference tables and requires PDF 1.4 or newer. It does not write incremental updates or linearised files.
- The reader is written to accept the shapes real producers emit — object streams, cross-reference streams, damaged tables, the six revisions of the standard security handler — and to say so in a warning when it has to repair something.

If your requirement is "merge these documents and tell me what you had to fix", that trade is a good one. If your requirement is a general PDF toolkit, Chapter 2 will tell you quickly whether this is the wrong library.

2. What SharpCorePDF Is Not

A precise boundary is part of the API. This chapter states what the library does not do, why, and what to use instead. If one of these lines is a requirement, SharpCorePDF is the wrong tool and no amount of configuration will change that.

It is not a PDF creation library

There is no API that starts from a blank page and draws text, images or vector graphics. Sources are PDF documents that already exist, supplied as a file, a byte array, a stream or a custom `IPdfContentSource`. The library can copy a page, select pages and combine documents; it cannot author content.

Use instead: a reporting or layout library that produces PDF, or a PDF toolkit with a content builder. SharpCorePDF can merge the reports afterwards.

It is not a PDF editor

There is no API to rotate a page, delete an annotation, replace text, fill a form, add a digital signature, redact content or change a page's media box. Page rotation and page size are *preserved* from the source; they are not editable through the merge plan.

Use instead: a viewer/editor SDK for interactive editing, or a PDF library that exposes page content modification. If you need to *assemble* a new document from existing pages, that is exactly what a merge plan does.

It is not a renderer, converter or OCR engine

The library does not rasterise pages, does not render them to an image, does not extract text, does not convert PDF to Word/HTML/text and does not recognise scanned characters.

Use instead: a rendering engine for thumbnails and previews, a text extraction library for search, and an OCR product for scans.

It is not a PDF/A or PDF/UA validator

The output is a well-formed PDF in the version you ask for. It is not validated against PDF/A, PDF/X, PDF/UA or any accessibility profile, and the library does not claim conformance. The primary source's catalog entries for language, marked content (`MarkInfo`) and output intents are copied, and colour profiles that belong to copied page resources travel with them, but no tagged structure tree is copied or rebuilt, and no profile is added, repaired or certified.

Use instead: a dedicated validator when a compliance profile is a contractual requirement.

It is not a general-purpose PDF parser

Inspection reports what a merge needs and what an operator wants to know before merging: the page count, page geometry, document information, outline size, embedded files, form presence, encryption state and warnings. It does not expose the object graph, does not enumerate fonts, does not report image resolutions and does not let you walk the content streams.

Use instead: a low-level PDF object model when you need to inspect arbitrary structures.

What is deliberately bounded inside a merge

These are not missing features in the sense above; they are decisions about what a merged document can sensibly contain. The library makes them explicit, and where a decision loses information it records a warning.

Area	Behaviour in 1.0.0
Document-level features	Metadata follows `MetadataPolicy` (the primary source by default); viewer preferences, page layout and mode, form fields, named destinations and embedded files come from the **primary source**, because they cannot sensibly exist twice in one document
Page-level features	Content, resources, annotations, page size and rotation are merged from **every** source
Outlines	Merged into one tree, top-level entries in source order; bookmarks whose destination cannot be resolved are skipped with a warning
Page labels	Dropped by default, because labels are tied to absolute page indices and misalign when sources are reordered or partially selected
AcroForm fields	The interactive form of the primary source is carried over; fields of other sources remain visible as page annotations but are not part of the merged form
Embedded files	Carried over from the primary source only
Named destinations	Carried over from the primary source; destinations in other sources are dropped with a warning
Encryption	Read: standard security handler revisions 2–6. Written: RC4-40, RC4-128, AES-128, AES-256. Public-key encryption is not supported
Output file structure	Classic cross-reference table, complete rewrite, PDF 1.4–2.0. No incremental update, no linearisation, no object streams in the output
Digital signatures	Signature fields and their widget annotations may be copied with the primary form or the pages, but a copied signature is not a valid signature of the merged document. The library does not sign and does not claim a copied signature remains trustworthy

> **The signature point deserves its own sentence.** If you merge a signed document, the result is a new document that no signer signed. Do not present the output as signed. Validate signatures before merging if their presence matters to your workflow.

What is deliberately bounded inside inspection

- Encrypted documents that cannot be opened are **described**, not thrown: `IsReadable` is `false`, `RequiresPassword` says whether a password would help, and inspection still returns a value. A merge of the same source throws `PdfEncryptionException` because it cannot proceed.
- Damaged documents are repaired where repair is safe and reported through warnings. When repair is not possible, inspection throws `PdfSourceFormatException` instead of returning a half-truth.
- Inspection reads the whole source into memory (bounded by `MaxSourceBytes`). It is a document-level operation, not a streaming scan.

Not a service, not a CLI

The repository contains a library and examples, not a hosted API, a command line tool or a Windows shell extension. A service that merges documents is a few dozen lines around `IPdfMerger`; the cookbook shows one.

The short version

If you need to…	SharpCorePDF
Combine existing PDFs, all pages or a selection	Yes
Reorder or repeat pages from a source	Yes
Preserve bookmarks, annotations and metadata	Yes
Encrypt the merged result	Yes

If you need to…	SharpCorePDF
Learn page counts, sizes, outline size or encryption state before merging	Yes
Create a PDF from scratch	No
Edit text, images or page geometry	No
Render, convert, OCR or extract text	No
Fill forms, sign or redact	No
Validate PDF/A or accessibility profiles	No

3. Installation and First Merge

This chapter takes an empty console project to a merged file and an inspection report. It should take about five minutes.

Install the package

```shell
dotnet add package SharpCorePDF --version 1.0.0
```

One package is enough. `SharpCorePDF` is a meta-package that brings the domain model, the application services, the engine and the dependency injection registration together. The four layer packages remain installable on their own — see Chapter 1 for when that is useful — but an application that merges documents does not need to choose.

The library targets **.NET 10** and **.NET 11**. A project on either target framework can reference it; the package selects the matching assembly.

Create the project

```shell
dotnet new console -n MergeDemo
cd MergeDemo
dotnet add package SharpCorePDF --version 1.0.0
```

Put two or more PDFs beside the project, or change the paths below. The example uses `cover.pdf`, `body.pdf` and `appendix.pdf`.

Register the services

The shortest correct wiring is a service collection. It costs four lines and gives you the merger, the inspector and the engine behind their ports, all as singletons.

```C#
using Microsoft.Extensions.DependencyInjection;
using SharpCorePDF.DependencyInjection;
using SharpCorePDF.Domain.Ports;

var services = new ServiceCollection()
    .AddSharpCorePDF()
    .BuildServiceProvider();

var merger = services.GetRequiredService<IPdfMerger>();
var inspector = services.GetRequiredService<IPdfInspector>();
```

`AddSharpCorePDF()` has three overloads: no arguments (default engine options), an explicit `PdfEngineOptions`, and an `IConfiguration` plus an optional section name. Chapter 10 covers all three and the options themselves.

You do not have to use dependency injection. The infrastructure types have public constructors, and Chapter 10 shows the manual wiring. The rest of this manual uses the container because it is the supported path for an application.

Merge

Add the following to `Program.cs`. The merge writes the selected pages of three documents, in the order they are added, to `merged.pdf`.

```C#
using SharpCorePDF.Application.Extensions;
using SharpCorePDF.Domain.Aggregates;
using SharpCorePDF.Infrastructure.Extensions;

var plan = MergePlan.Create()
    .AddFile("cover.pdf")
    .AddFile("body.pdf", pages: "1-3,7")
    .AddFile("appendix.pdf", displayName: "Appendix", pages: "2-");

var result = await merger.MergeToFileAsync(plan, "merged.pdf");

Console.WriteLine($"{result.PageCount} pages, {result.OutputBytes} bytes, {result.Duration.TotalMilliseconds:F0}
ms");
Console.WriteLine($"Primary source: {plan.FindSource(result.PrimarySourceId)?.DisplayName}");
```

Run it:

```SHELL
dotnet run
```

Expected output, with different timings and sizes:

```TEXT
6 pages, 184213 bytes, 37 ms
Primary source: cover.pdf
```

Three things are worth noticing in that snippet.

The page selection is a string. `"1-3,7"` means pages 1, 2, 3 and 7, in that order. `"2-"` means page 2 to the end. The notation is explained in Chapter 5; `null` or an omitted argument means every page.

The display name is for humans. It defaults to the file name and appears in `MergeResult`, in the source summaries and in messages. It does not have to be unique and it does not affect the output.

The extension methods come from two namespaces. `AddFile`, `AddBytes` and `AddStream` live in `SharpCorePDF.Infrastructure.Extensions`; `MergeToFileAsync` and `MergeToBytesAsync` live in `SharpCorePDF.Application.Extensions`. A missing `using` for either produces a compiler error that mentions the receiver type, not the namespace.

Read the result

`MergeResult` is the record of what happened. The properties used most often are:

Property	Meaning
`PageCount`	Pages written to the output
`SourceCount`	Number of sources that contributed at least one page
`OutputBytes`	Size of the merged document
`OutputVersion`	The PDF version written in the header
`Duration`	Wall-clock time spent in the merge
`PrimarySourceId`	The source that supplied document-level features
`Sources`	One `MergedSourceSummary` per input, in plan order, with contributed page count
`Warnings`	Every warning recorded while reading, merging or writing

The destination is not repeated in the result: the caller already has the path or the stream it passed in. Use `plan.FindSource(result.PrimarySourceId)` when you need the display name of the primary source, as in the example

above.

If `Warnings` is empty, nothing had to be repaired or dropped. If it is not empty, the merge still succeeded; Chapter 9 tells you how to triage the warning kinds.

Inspect before you merge

Inspection uses the same reader. Use it to check a document before adding it to a plan, or to display a summary to a user.

```C#
using SharpCorePDF.Infrastructure.Sources;

var info = await inspector.InspectAsync(new PdfFileSource("body.pdf"));

Console.WriteLine($"{info.PageCount} pages, PDF {info.Version}, encrypted: {info.IsEncrypted}");
Console.WriteLine($"Readable: {info.IsReadable}, warnings: {info.Warnings.Count}");

foreach (var page in info.Pages.Take(3))
{
    Console.WriteLine(
        $"  page {page.Number}: {page.Size.WidthPoints:F0} x {page.Size.HeightPoints:F0} pt, " +
        $"rotation {page.Rotation.Degrees}");
}
```

`PdfFileSource` is a small `IPdfContentSource` that opens the file when it is read. `InspectAsync` has an optional password parameter — `InspectAsync(source, password, cancellationToken)` — which Chapter 8 covers together with the rest of the read models.

A complete `Program.cs`

The pieces above, assembled and made a little more careful: it validates the plan before doing any work, prints warnings, and returns a non-zero exit code if the merge fails.

```C#
using Microsoft.Extensions.DependencyInjection;
using SharpCorePDF.Application.Extensions;
using SharpCorePDF.Domain.Aggregates;
using SharpCorePDF.Domain.Exceptions;
using SharpCorePDF.Domain.Ports;
using SharpCorePDF.DependencyInjection;
using SharpCorePDF.Infrastructure.Extensions;

var services = new ServiceCollection().AddSharpCorePDF().BuildServiceProvider();
var merger = services.GetRequiredService<IPdfMerger>();

var plan = MergePlan.Create()
    .AddFile("cover.pdf")
    .AddFile("body.pdf", pages: "1-3,7")
    .AddFile("appendix.pdf", displayName: "Appendix");

try
{
    plan.EnsureValid();

    var result = await merger.MergeToFileAsync(plan, "merged.pdf");
    Console.WriteLine($"Wrote {result.PageCount} pages, {result.OutputVersion}.");

    foreach (var warning in result.Warnings)
    {
        Console.WriteLine($"warning {warning.Kind}: {warning.Message}");
    }

    return 0;
}
catch (PdfException exception)
```

```
{
    Console.Error.WriteLine($"{exception.GetType().Name}: {exception.Message}");
    return 1;
}
```

Two details in that listing are deliberate. `EnsureValid()` runs before the destination file is created, so an invalid plan cannot leave a truncated file behind. And the `catch` is for `PdfException`, the base type of every library failure, rather than `Exception`; an `OperationCanceledException` or an `OutOfMemoryException` is not a PDF problem and should not be reported as one.

Where to go next

- To control which pages come from which document, read Chapter 5.
- To set metadata, version, bookmarks or encryption, read Chapters 6 and 7.
- To decide what to do when a source is damaged or password-protected, read Chapters 8 and 9.
- To wire the library into an existing application, read Chapter 10.

4. The Merge Plan

Everything the library does with several documents is described by a `MergePlan`: an ordered list of sources and one set of output options. A plan is a value. It is not a builder that gets mutated, not a job that gets executed, and not a handle that has to be disposed.

```C#
var plan = MergePlan.Create()
    .AddFile("cover.pdf")
    .AddFile("chapter-1.pdf", pages: "1-")
    .AddFile("chapter-2.pdf", pages: "1-")
    .WithOutput(options => options with { TargetVersion = PdfVersion.V1_7 });
```

That single expression states the order of the merge, what each source contributes and how the output must be written. The next chapters unpack the three parts; this one explains how they fit together.

Building a plan

`MergePlan.Create()` starts an empty plan. Every method that changes it returns a new plan, so the usual style is to chain calls and keep the result.

Member	Purpose
`Create(MergeOutputOptions? output = null)`	Starts a plan, optionally with output options
`For(params PdfSource[] sources)`	Starts a plan from sources you built yourself
`AddSource(PdfSource source)`	Appends a source entity
`AddSources(IEnumerable<PdfSource> sources)`	Appends several sources, in enumeration order
`RemoveSource(PdfSourceId id)`	Returns a plan without the source with that identity
`WithOutput(MergeOutputOptions output)`	Replaces the output options
`WithOutput(Func<MergeOutputOptions, MergeOutputOptions> mutate)`	Adjusts them with a `with` expression
`WithPrimarySource(PdfSourceId? id)`	Sets, or clears, the primary source
`FindSource(PdfSourceId id)`	Returns the source, or `null`
`Validate()`	Returns every problem that can be found without reading a document
`EnsureValid()`	Throws `MergePlanValidationException` if there is one
`Sources`	The sources, in merge order
`PrimarySource`	The source that supplies document-level features
`SourceCount`	The number of sources

The convenience methods `AddFile`, `AddBytes`, `AddStream` and `AddContent` are extension methods from `SharpCorePDF.Infrastructure.Extensions`. They create the matching content source and call `AddSource`, and they accept the page selection in its textual notation. The core `AddSource` is enough when you already have a `PdfSource`.

Sources and identity

A `PdfSource` has four pieces of state:

- **`Content`** — an `IPdfContentSource`, the object that can open the document for reading.

- `DisplayName` — the name used in messages and in `MergeResult`. It defaults to the content source's own name and can be changed with `Rename`.
- `Pages` — a `PageSelection`; `PageSelection.All` when nothing was specified.
- `Password` — the user or owner password for an encrypted document, or `null`.

It also has an `Id`, a `PdfSourceId` generated when the source is created. The identity is what lets a plan refer to a source later — for example as the primary source — without relying on its position, and it is what validation uses to detect an accidental duplicate.

Two `AddFile` calls for the same path produce two sources with two identities. That is legal and is how you repeat a document:

```C#
var plan = MergePlan.Create()
    .AddFile("terms.pdf", displayName: "Terms (before)")
    .AddFile("body.pdf")
    .AddFile("terms.pdf", displayName: "Terms (after)");
```

The duplicate that SharpCorePDF rejects is the same *identity* appearing twice, which can only happen if you pass one `PdfSource` instance to `AddSource` twice or supply an explicit identity through `PdfSource.Create(id, content, …)`. `AddSource` throws `MergePlanValidationException` immediately; it is a programming error rather than a use case, and it fails before any document is read.

Sources also support non-mutating edits of their own:

```C#
var source = PdfSource.Create(new PdfFileSource("report.pdf"))
    .WithPages(PageSelection.FromTo(2, 5))
    .Rename("Report, pages 2 to 5");
```

`WithPages`, `WithPassword` and `Rename` keep the identity, so a plan that already refers to the source as its primary source keeps working after the edit.

Order is the merge

The order of `Sources` is the order of the output. There is no sorting, no grouping by path and no implicit title page. Pages contributed by the first source come first, then those of the second, and so on. Within one source, the page selection decides the order, which is why `"3,1"` is a legal selection and produces page 3 before page 1.

This makes a plan readable as a table of contents. If you need a different order, build a different plan; reordering is intentional, not an optimisation the library performs behind your back.

The primary source

Some things can only exist once in a PDF: the document information dictionary, viewer preferences, page layout and mode, the interactive form, named destinations, embedded files. When sources disagree, the merge must choose one of them. It chooses the **primary source**.

By default the primary source is the first source in the plan. `WithPrimarySource` changes it:

```C#
var cover = plan.Sources[0];
var body = plan.Sources[1];

var withBodyMetadata = plan.WithPrimarySource(body.Id);
```

The primary source is usually the one whose identity matters most — the cover whose title should survive, or the document whose form fields the user expects to keep working. It does not have to be the first source, and setting it does not change the order of the pages. Chapter 6 lists exactly which output options follow the primary source and which can override it.

`PrimarySource` throws `MergePlanValidationException` for an empty plan, and for an id that is not part of the plan. Both are also caught by `Validate`.

Validation happens twice

`Validate` checks what can be known without opening a document:

- the plan has at least one source;
- no two sources share an identity;
- every source has a non-empty display name;
- the output options are internally consistent — a supported target version, a known encryption algorithm, a primary source that belongs to the plan.

`MergeAsync` calls `EnsureValid` itself, so a plan that was never validated still cannot reach the engine. Validation then continues against the real documents: each source is opened, its page count is known, and a selection such as `"9-12"` against a four-page document fails with `MergePlanValidationException` before the output is written.

Calling `EnsureValid` yourself is still worthwhile when you want the failure to happen before you create a destination file, or when you want to report all problems at once:

```C#
var problems = plan.Validate();

if (problems.Count > 0)
{
    throw new MergePlanValidationException(problems);
}
```

The exception's `Errors` property preserves the individual messages. They are written for humans and are not part of the compatibility contract; do not parse them.

Executing a plan

There are four ways to run a plan, all of them thin wrappers over the same engine:

Call	Destination	Returns
`merger.MergeAsync(plan, stream, ct)`	A writable stream you own	`MergeResult`
`merger.MergeToFileAsync(plan, path, ct)`	A file, created or overwritten	`MergeResult`
`merger.MergeToBytesAsync(plan, ct)`	An in-memory buffer	`byte[]`
`merger.MergeAsync(sources, stream, ct)`	A stream, from a list of content sources	`MergeResult`

`MergeToFileAsync` creates the destination directory when it does not exist, opens the file with `FileMode.Create` and exclusive sharing, and lets the merge write into it. The stream overload flushes the destination but does not close it: the caller owns the stream. `MergeToBytesAsync` is convenient for an HTTP response, but it holds the whole result in memory; use a file or a stream for large documents.

The list overload is a shorthand when every page of every source is wanted:

```C#
var result = await merger.MergeAsync(
    new IPdfContentSource[]
    {
        new PdfFileSource("a.pdf"),
        new PdfFileSource("b.pdf"),
    },
    destination);
```

It builds a plan with default options. As soon as you need a page selection, a primary source or non-default output options, build the plan explicitly; the shorthand does not grow parameters for those.

> **Cancellation leaves a partial destination.** A cancelled `MergeToFileAsync` throws `OperationCanceledException`, and the file it was writing may be incomplete. If the destination must never be observed half-written, merge to a temporary file in the same directory and move it into place after the call succeeds. Chapter 12 shows the pattern.

Reuse and thread safety

A plan never changes after it is built, so it can be stored, compared, cached and shared. The built-in content sources open a fresh read for every execution: `PdfFileSource` opens the file again, `PdfByteArraySource` returns a new read-only `MemoryStream` over the same bytes, and `PdfStreamSource` captures the stream once, under a lock, and serves every later read from the captured bytes. A plan whose sources are one of these can be executed concurrently.

A custom `IPdfContentSource` must honour the same contract — every call to `OpenReadAsync` returns an independent readable stream — if you intend to reuse or share its plan. If it cannot, create a new source, and therefore a new plan, per execution.

The merger itself is thread safe. The container registers it as a singleton, and the engine keeps no per-merge state on the instance.

5. Page Selection

A page selection says which pages of one source document enter the merged document, and in what order. It is the only place where the library lets you rearrange pages, and it is deliberately small: a comma-separated list of one-based ranges.

```C#
var plan = MergePlan.Create()
    .AddFile("report.pdf", pages: "1,3-5,8-")
    .AddFile("appendix.pdf", pages: "2,1");
```

The notation

Text	Meaning	Pages contributed from a 10-page source
`"1"`	One page	1
`"2-4"`	A closed range, inclusive	2, 3, 4
`"3-"`	From a page to the last page	3, 4, 5, 6, 7, 8, 9, 10
`"-2"`	From the first page to a page	1, 2
`"1,3-5,8-"`	Several ranges, in this order	1, 3, 4, 5, 8, 9, 10
`"3,1,3"`	Out of order, with a duplicate	3, 1
`" 2 - 4 "`	Whitespace around numbers is ignored	2, 3, 4

The rules behind the table:

- **Pages are one-based.** Page 1 is the first page a reader shows. PDF's internal zero-based index never appears in this API.
- **Ranges are inclusive at both ends.** `2-4` contains 2, 3 and 4.
- **A range must not run backwards.** `5-2` is a `FormatException`, not an empty selection.
- **A range must start at page 1 or later.** `0-3` is a `FormatException`.
- **Order is preserved.** Ranges are read left to right, and pages inside a range ascend. `"3,1"` puts page 3 before page 1.
- **Duplicates collapse on first occurrence.** `"3,1,3"` selects page 3, then page 1; the second 3 is dropped because it was already selected. Repeating a page is not how you duplicate it — add the source twice instead (Chapter 4).
- **A closed range may pass the last page; an open range may not start past it.** `"2-99"` on a four-page document contributes pages 2, 3 and 4 to the merge, and `"9-"` fails because it starts after the last page. `Resolve` itself still returns 2 to 99, so a preview must drop numbers greater than `PageCount`; the merge does that when it copies pages. A closed range that starts inside the document never throws for its tail.
- **The selection counts physical pages, not page labels.** If a document displays "iv" on its fourth physical page, select `"4"`, not `"-iv"`. Page labels are a presentation feature and are dropped by default anyway (Chapter 6).

Whitespace around the whole text and around each number is accepted. An empty string is not a valid selection: `PageSelection.Parse("")` throws `FormatException`. The `AddFile` extension treats `null`, `""` and whitespace as "every page" because it checks before calling `Parse`; if you call `PageSelection.Parse` yourself, handle the empty case explicitly.

Building a selection without text

Text is a convenience. The value objects are public, and a selection can be built directly when it comes from a UI, a database or a calculation.

```C#
using SharpCorePDF.Domain.Entities;
using SharpCorePDF.Domain.ValueObjects;
using SharpCorePDF.Infrastructure.Sources;

PageSelection firstThree = PageSelection.FromTo(1, 3);
PageSelection appendix = PageSelection.FromPage(12);
PageSelection coverAndBack = PageSelection.Of(
    PageRange.Single(1),
    PageRange.ToPage(2),
    PageRange.FromTo(8, 9));
PageSelection everything = PageSelection.All;

var source = PdfSource.Create(new PdfFileSource("report.pdf"), pages: coverAndBack);
```

Factory	Produces
`PageSelection.All`	`1-`
`PageSelection.Single(page)`	`page`
`PageSelection.FromTo(from, to)`	`from-to`
`PageSelection.FromPage(from)`	`from-`
`PageSelection.Of(params PageRange[])`	The ranges in the given order
`PageRange.Single(page)`	One range of one page
`PageRange.FromTo(from, to)`	A closed range
`PageRange.FromPage(from)`	An open-ended range
`PageRange.ToPage(to)`	A range from page 1
`PageRange.All`	`1-`
`PageSelection.Parse(text)` / `TryParse`	A selection from the notation
`PageRange.Parse(text)` / `TryParse`	A single range from the notation

`PageSelection` is immutable. `Add` appends a range and `Concat` appends another selection, both returning a new instance. `Ranges` exposes the ranges, `RangeCount` their number and `IsAll` tells you whether the selection is the `1-` shortcut.

Resolving against a document

A selection is document-independent until it is resolved. `Resolve(pageCount, sourceName)` returns the concrete page numbers, in merge order and without duplicates, and throws `MergePlanValidationException` when the document has no pages or a range starts past its end.

```C#
var info = await inspector.InspectAsync(new PdfFileSource("report.pdf"));
var selection = PageSelection.Parse("1,3-5,8-");

IReadOnlyList<int> pages = selection.Resolve(info.PageCount, info.SourceName);

Console.WriteLine(string.Join(", ", pages));
```

You rarely call `Resolve` in application code. The merger resolves every selection while it prepares the merge, before the engine writes pages, and the same resolution rules apply. Calling it yourself is useful for a preview — "this plan will produce 14 pages" — or for rejecting a request before a destination file exists.

Two properties are worth knowing when you build that preview:

- `ResolveCount(pageCount, sourceName)` returns `Resolve(...).Count`. It counts the *selected numbers*, including a closed range's tail beyond the document, so it can overstate what is written. For an exact count, count only the resolved numbers at most `pageCount`, or use the `PageCount` of a completed merge.
- Selection is not an I/O optimisation. The reader still reads the whole source document; the selection decides which pages are copied into the result, not how much of the file is read.

Preflight instead of exceptions

When page ranges come from a user, checking them before the merge produces a better message than catching the exception afterwards. The inspection model makes that a few lines:

```C#
var info = await inspector.InspectAsync(new PdfFileSource(path));

if (!info.IsReadable)
{
    return $"'{path}' cannot be read: {info.Warnings.FirstOrDefault()?.Message}";
}

if (PageSelection.TryParse(requestedPages, out var selection))
{
    try
    {
        var pages = selection.Resolve(info.PageCount, info.SourceName);
        Console.WriteLine($"{pages.Count} of {info.PageCount} pages selected.");
    }
    catch (MergePlanValidationException exception)
    {
        Console.WriteLine(exception.Message);
    }
}
```

The preflight and the merge share the same `Resolve` implementation, so an accepted selection is accepted by the merge. That is the advantage of the value object being public: you do not have to reimplement its rules in a validator.

Common shapes

```C#
// Everything, explicitly.
.AddFile("a.pdf", pages: "1-")

// Everything except the cover, then the cover at the end.
.AddFile("report.pdf", pages: "2-")
.AddFile("report.pdf", pages: "1", displayName: "Cover")

// A chapter per source, each starting at its own page 1.
.AddFile("chapter-01.pdf", pages: "1-")
.AddFile("chapter-02.pdf", pages: "1-")
.AddFile("chapter-03.pdf", pages: "1-")

// Two specific pages in reverse order.
.AddFile("spread.pdf", pages: "2,1")
```

The third shape is the one to reach for when documents are already split into chapters: the merge plan mirrors the file list, and no page arithmetic is needed. The second shape shows that a document may appear twice; each appearance is a separate source with its own selection and display name.

A note on bookmarks and links

When a document is partially selected, bookmarks, links and form fields that point at pages outside the selection still need a destination. The library points them at the first page the source contributes instead of dropping the whole bookmark or link. The result opens, the navigation stays usable, and the destination is approximate by design.

If a bookmark's destination cannot be resolved at all — for example because it refers to an object the source itself does not contain — it is skipped and a warning is recorded. Chapter 9 lists the warning kinds that can appear for bookmarks, named destinations and page labels.

6. Output Options

`MergeOutputOptions` describes the document you want, as opposed to the pages you want. It is an immutable record with a dozen settings, all optional, and it is attached to a plan with `WithOutput`:

```C#
var now = new DateTimeOffset(2025, 12, 31, 12, 0, 0, TimeSpan.Zero);

var plan = MergePlan.Create()
    .AddFile("cover.pdf")
    .AddFile("body.pdf")
    .WithOutput(options => options with
    {
        MetadataPolicy = MetadataPolicy.Merge,
        MetadataOverride = new PdfMetadata { Title = "Annual report 2025" },
        ModificationDate = now,
    });
```

Setting	Type	Default	Effect
`TargetVersion`	`PdfVersion`	`V1_7`	The version written to the file header
`PrimarySourceId`	`PdfSourceId?`	`null` (first source)	Which source leads document-level features
`MetadataPolicy`	`MetadataPolicy`	`FromPrimarySource`	Where document information comes from
`MetadataOverride`	`PdfMetadata?`	`null`	Fields that replace whatever the policy produced
`ModificationDate`	`DateTimeOffset?`	`null` (now)	The modification date, and the basis of the file identifier
`OutlinePolicy`	`OutlinePolicy`	`Preserve`	Whether bookmarks are merged or dropped
`PageLabelPolicy`	`PageLabelPolicy`	`Drop`	Whether page labels are copied or dropped
`PreserveAnnotations`	`bool`	`true`	Whether page annotations are copied
`PreserveAcroForm`	`bool`	`true`	Whether the primary source's interactive form is copied
`PreserveEmbeddedFiles`	`bool`	`true`	Whether the primary source's attachments are copied
`Encryption`	`PdfEncryptionOptions?`	`null`	Whether the result is encrypted
`Producer`	`string`	`"SharpCorePDF <version>"`	The value of the `/Producer` entry

The defaults follow one principle: **lose as little as possible, without inventing anything.** Pages, annotations and bookmarks are merged from every source. Features that cannot meaningfully exist twice in one document — document information, the interactive form, attachments, page labels — are taken from one source, the primary one. Nothing is encrypted unless you ask for it.

`MergeOutputOptions.Validate()` returns a list of problems rather than throwing, and `MergePlan.Validate()` includes those problems in its own report, so a bad combination surfaces before a file is opened. The checks cover an unsupported target version, undefined policy values, a blank producer and encryption settings that the target version cannot express (Chapter 7).

Target version

The engine writes a classic cross-reference table, which requires PDF 1.4 or newer; `V1_4` through `V2_0` are accepted and anything outside that range is rejected. `V1_7` is the default because it is the most widely supported version that still uses the traditional file structure.

The target version is a claim, not a conversion. The library does not rewrite content to match an older version and does not verify that copied objects only use features of that version. If your sources use transparency, layers or other features introduced after 1.4 and you write a 1.4 file, the file may open in lenient viewers and fail in strict ones. When in doubt, keep the default, or set the target to the highest version among your sources.

Choosing the primary source

`PrimarySourceId` names the source that leads document-level features. By default it is the first source in the plan, but any source may be promoted:

```C#
var body = PdfSource.Create(new PdfFileSource("body.pdf"), displayName: "Body");
var cover = PdfSource.Create(new PdfFileSource("cover.pdf"), displayName: "Cover");

var plan = MergePlan.Create()
    .AddSource(cover)
    .AddSource(body)
    .WithPrimarySource(body.Id);   // metadata, form and attachments come from body.pdf

plan.EnsureValid();
```

`WithPrimarySource` is the shortcut on the plan; `PrimarySourceId` in the output options is the same setting, useful when the options are built elsewhere. A primary source must exist in the plan and must be identified by `PdfSourceId`, not by file path. The call made last wins: `WithPrimarySource` changes the current output options, and a later `WithOutput` can replace the value again.

The primary source is only consulted for features that cannot be merged. Its pages are not treated specially: page order still comes from the plan, and its bookmarks are still merged with the rest. Every source in the plan must contribute at least one page, so a primary source whose selection resolves to no pages is rejected with `MergePlanValidationException` before anything is written; the primary is always a source that actually contributes pages.

Document information

`MetadataPolicy` has three values:

Value	Result
`FromPrimarySource`	The primary source's document information, copied including custom entries
`Merge`	The primary source's values, with empty fields filled from the other sources in plan order
`None`	No document information, apart from the producer and modification date

`FromPrimarySource` is the default because a merge is usually judged by its pages, and the metadata of the document that leads the merge is the most defensible choice. `Merge` is useful when the first file is a bare cover with no title but a later file has one. `None` produces the smallest, least revealing dictionary.

`MetadataOverride` is applied after the policy and replaces only the fields you set:

```C#
MetadataOverride = new PdfMetadata
{
    Title = "Annual report 2025",
    Author = "Finance",
    Keywords = "annual, report, 2025",
    CustomProperties = new Dictionary<string, string>
    {
        ["department"] = "finance",
    },
},
```

An empty string is a value, not an absence: it replaces the field with an empty string. `null` leaves the field as the policy produced it. Custom properties are merged by key, with the override winning. `Producer` and `ModificationDate` are the two fields `MetadataOverride` cannot set: the producer is always `MergeOutputOptions.Producer`, and the modification date is always `MergeOutputOptions.ModificationDate` (or the current UTC time when it is `null`). Setting either one in the override has no effect; set the output option instead.

`ModificationDate` describes the merge run, not the source document, so it is never taken from the policy or from `MetadataOverride`. Set it on the output options to fix the date for a whole plan.

A PDF can carry a second kind of metadata: an XMP packet, a small XML document stored in a stream rather than in the information dictionary. When the metadata policy is not `None`, the primary source's XMP packet is copied unchanged. The library does not regenerate it, so an XMP reader may see the primary source's title even after `MetadataOverride` changes the title in the information dictionary. If consistent XMP matters to your consumers, strip or rewrite it downstream.

Bookmarks, labels and annotations

These settings are independent, and turning one off does not turn off the others.

```C#
.WithOutput(options => options with
{
    OutlinePolicy = OutlinePolicy.Drop,       // no bookmarks at all
    PageLabelPolicy = PageLabelPolicy.FromPrimarySource,
    PreserveAnnotations = false,              // no links, highlights, stamps or widgets
    PreserveAcroForm = false,                 // no interactive form
    PreserveEmbeddedFiles = false,            // no attachments
})
```

Bookmarks. `Preserve` merges the outline trees of every source into one tree whose top-level items follow source order. Bookmarks are remapped to their page's new position; a bookmark that points at a page outside the source's selection is pointed at the first selected page instead. `Drop` writes no outline at all and is the right choice when source bookmarks are meaningless in the new document.

Page labels. Page labels are the displayed page numbers — roman numerals for a preface, "A-1" for an appendix. They are tied to absolute page indices, so they misalign as soon as sources are reordered or partially selected, which is why `Drop` is the default. `FromPrimarySource` copies the primary source's labels and applies them from the first output page, which is only accurate when the primary source starts at page 1 of the result.

Annotations. `PreserveAnnotations` controls every annotation on the merged pages: link annotations, highlights, notes, stamps, and the widget annotations that draw form fields. Turning it off is common for print-oriented output. Bookmarks are not annotations and are unaffected; the cost is that internal links stop working.

Forms. `PreserveAcroForm` copies the interactive form of the primary source only. Form fields from the other sources are not added to the form, but their widget annotations remain on the pages when annotations are preserved, so they stay visible without being fillable. Merging two independent forms into one is not attempted: field names would collide and the result would be ambiguous.

Attachments. `PreserveEmbeddedFiles` copies the embedded files of the primary source. Attachments from the other sources are not merged; if several sources carry attachments, choose the source with the complete set as the primary, or attach files to the result afterwards.

Reproducible output

`ModificationDate` does two jobs. It is written as the modification date, and it is the input from which the document identifier is derived. Supply a fixed value and, for unencrypted output, the same inputs produce the same bytes on every run. Leave it `null` and the current UTC time is used, so every run differs.

Reproducibility has limits worth stating plainly. Encrypted output uses fresh random material, so two encrypted merges of the same input are not byte-identical even with a fixed date (Chapter 7). The producer string contains the assembly version, so upgrading the library changes the bytes. And reproducible bytes are not a signature: they show that two files were produced the same way, not that the content is trustworthy.

```C#
var options = MergeOutputOptions.Default with
{
    ModificationDate = new DateTimeOffset(2025, 1, 1, 0, 0, 0, TimeSpan.Zero),
    Producer = "Contoso Report Pipeline",
};
```

7. Encryption and Permissions

A merged document is written in the clear unless you set `Encryption`. That is true even when every source was encrypted: a merge is a new document, and the passwords of its sources are not carried into it. Encryption is therefore a decision about the output, made once, with `PdfEncryptionOptions`.

```C#
var encryption = new PdfEncryptionOptions
{
    Algorithm = PdfEncryptionAlgorithm.AesWith256BitKey,
    UserPassword = "reader",        // opens the document with the permissions below
    OwnerPassword = "publisher",    // opens it with every permission
    Permissions = PdfPermissions.PrintOnly,
};

var plan = MergePlan.Create()
    .AddFile("report.pdf", password: "source-password")
    .WithOutput(options => options with { Encryption = encryption });

plan.EnsureValid();
```

Setting	Type	Default	Meaning
`Algorithm`	`PdfEncryptionAlgorithm`	`AesWith256BitKey`	The cryptosystem written into the security dictionary
`UserPassword`	`string`	`""`	Opens the document with `Permissions`
`OwnerPassword`	`string`	`""`	Opens it with every permission, ignoring `Permissions`
`Permissions`	`PdfPermissions`	`All`	What a reader who supplied the user password may do

Algorithms

Value	Security handler	Key	Minimum target version	Notes
`Rc4With40BitKey`	Revision 2	40-bit RC4	1.4 (any allowed)	Legacy compatibility only; not protection
`Rc4With128BitKey`	Revision 3	128-bit RC4	1.4 (any allowed)	Legacy; RC4 is broken
`AesWith128BitKey`	Revision 4	128-bit AES	1.5	AESV2 crypt filter
`AesWith256BitKey`	Revision 6	256-bit AES	1.7	AESV3 crypt filter, the default

`AesWith256BitKey` is the default for the same reason it is the default almost everywhere: AES-256 with the hardened revision 6 password hash is the strongest option, and every current reader supports it. Revision 5 of the standard security handler is deliberately not offered, because its single-pass SHA-256 password hash was replaced by revision 6 precisely because it was too cheap to attack. The RC4 algorithms exist for workflows that include a tool which cannot go beyond them; do not choose them for new documents.

The algorithm and the target version must agree. `Validate(TargetVersion)` reports AES-128 below PDF 1.5 and AES-256 below PDF 1.7, and `MergePlan.Validate()` includes those errors, so the mismatch is caught before anything is written rather than producing a file that readers reject.

Passwords

The two passwords have different jobs:

- The **user password** opens the document and is subject to `Permissions`.
- The **owner password** opens the document with every permission, whatever `Permissions` says.

The rules that follow from those jobs:

- **Empty user password.** The document opens without a prompt. This is the shape of a document that is only restricted against editing: anyone can read it, and only the owner password lifts the restrictions. A cooperating viewer still honours `Permissions` for the reader who did not supply the owner password.
- **Empty owner password.** The user password becomes the owner password as well. A document never needs more credentials than it was written with, and there is no separate owner secret to lose.
- **Both empty.** Encryption is still applied, with a key derived from empty passwords. Anyone can derive that key from the file, so the document is effectively unprotected. The library allows it because the difference between "no encryption" and "meaningless encryption" can matter to a downstream tool; it is not a sensible choice for a real document.
- **Truncation.** Each revision defines a maximum password length — 32 bytes for revisions 2 to 4 and 127 bytes for revision 6. A longer password is truncated by the format. `Validate` rejects a password whose UTF-8 encoding exceeds the limit, so you learn about the truncation at validation time instead of discovering that a shorter password opens the file.
- **Encoding.** Revisions 2 to 4 predate an agreement on how passwords are encoded. A password that survives a Latin-1 round trip is written as Latin-1 and anything else as UTF-8, which is what Adobe, qpdf and pypdf hash as well. Revisions 5 and 6 are UTF-8 by definition. SASL preparation is not applied, matching what readers accept.

Both passwords are checked by validation, so a password that is too long for the chosen algorithm is reported as a validation problem before the merge starts, rather than silently truncated.

Permissions

`PdfPermissions` is a flags enumeration whose values are the permission bits of the PDF encryption dictionary. Combine them with `|`.

Flag	Allows
`Print`	Printing
`PrintHighQuality`	Printing at a quality that reproduces the content faithfully
`ModifyContents`	Changing the pages and the document as a whole
`CopyContent`	Copying text and graphics out
`Annotate`	Adding and changing annotations, and filling in form fields
`FillFormFields`	Filling in form fields even when `Annotate` is not allowed
`ExtractForAccessibility`	Screen readers and similar tools extracting content even without `CopyContent`
`Assemble`	Inserting, rotating and deleting pages, and building bookmarks and thumbnails
`All`	Every flag above — a document that only needs a password to open
`PrintOnly`	`Print | ExtractForAccessibility` — readable and printable, nothing else
`None`	Nothing beyond viewing the pages

Permissions are **advisory, not enforcement**. They tell a cooperating viewer what to offer. A reader that ignores them, a tool that parses the file directly, or anyone with the owner password is not stopped. PDF encryption protects the bytes of strings and streams; it does not implement digital rights management, and it should not be described to users as if it did. For documents that must not be copied, the realistic control is who receives the file.

Two smaller consequences of the flags:

- Revision 2 does not define `FillFormFields` or the flags after it. They are still written, but a reader that predates them ignores those bits, so the extra actions you allowed are simply not offered. If you target the oldest readers, restrict yourself to `Print`, `ModifyContents`, `CopyContent` and `Annotate`.
- An empty user password with `PrintOnly` is the usual shape of a "read and print, do not edit" document: no password to open, an owner password for anyone who needs to change it, and a viewer that offers only the permitted actions.

Encrypted sources

The encryption of a source and the encryption of the result are unrelated. A source is opened with the password given to `PdfSource.Create`, `AddFile`, `AddBytes` or `AddStream`; that password may be either the user or the owner password, and `null` is correct for a document that needs none.

```C#
var plan = MergePlan.Create()
    .AddFile("confidential.pdf", password: "reader")
    .AddFile("public.pdf")
    .WithOutput(options => options with
    {
        Encryption = new PdfEncryptionOptions
        {
            UserPassword = "reader-2025",
            OwnerPassword = "publisher-2025",
            Permissions = PdfPermissions.Print | PdfPermissions.CopyContent,
        },
    });
```

Decryption happens while the source is read. The result is written from the parsed object model and then encrypted with the new settings, so the source passwords leave no trace in the output and do not have to be reused. It also means the result inherits nothing: an encrypted source merged without `Encryption` produces a clear document, and a clear source merged with `Encryption` produces an encrypted one.

When a source cannot be opened, the merger throws `PdfEncryptionException` with `RequiresPassword` telling you which problem you have:

- `RequiresPassword == true` — the password was missing or wrong. Ask for one and retry.
- `RequiresPassword == false` — the document uses a security handler the engine does not implement, or its encryption dictionary is malformed. No password will help.

Inspection never throws for an encrypted document (Chapter 8). It reports `RequiresPassword` and `SecurityHandlerRevision` on `PdfDocumentInfo` instead, which is how a caller can ask for a password before attempting a merge:

```C#
var info = await inspector.InspectAsync(new PdfFileSource("confidential.pdf"));

if (info.RequiresPassword)
{
    Console.WriteLine($"'{info.SourceName}' needs a password.");
}
```

Reproducibility and encryption

Encryption draws fresh random material for the file key, the password salts and every initialization vector. Two encrypted merges of identical input with identical settings are therefore not byte-identical, even with a fixed `ModificationDate` (Chapter 6). They are equivalent documents with different bytes; compare their content, not their hashes.

This is the expected behaviour, not a defect: reusing an initialization vector or a salt across documents would weaken the encryption. If you need reproducible bytes for caching or auditing, produce the unencrypted document reproducibly and encrypt it in a separate step whose output you do not compare byte for byte.

8. Inspection

`IPdfInspector` answers "what is in this file?" without changing it and without producing a merge. It is the read side of the library, and it uses the same reader as the merger, so what you inspect is what you will merge.

```C#
var info = await inspector.InspectAsync(
    new PdfFileSource("report.pdf"),
    password: null,
    cancellationToken: cancellationToken);
```

The method takes any `IPdfContentSource`, not just a file, so the same call works for a byte array or a stream. There is no inspection overload that takes a path directly; wrap the path in a `PdfFileSource`, which also supplies the display name used in messages.

The document model

`PdfDocumentInfo` is an immutable record. The properties you are most likely to use:

Property	Type	Meaning
`SourceName`	`string`	The display name of the source
`Version`	`PdfVersion`	The version from the document header
`PageCount`	`int`	Number of pages, or `0` when the page tree is unreadable
`IsReadable`	`bool`	Whether the structure could be read at all
`IsEncrypted`	`bool`	Whether the document declares encryption
`RequiresPassword`	`bool`	Whether a password is what stands in the way
`SecurityHandlerRevision`	`int?`	The revision of the security handler, or `null`
`Metadata`	`PdfMetadata`	Title, author, subject, keywords, dates and custom entries
`Pages`	`IReadOnlyList<PdfPageInfo>`	One entry per page, in document order
`HasOutlines`	`bool`	Whether the document has bookmarks
`OutlineItemCount`	`int`	Bookmarks, counting nested items
`EmbeddedFileCount`	`int`	Number of attachments
`HasAcroForm`	`bool`	Whether the document has an interactive form
`SourceLengthBytes`	`long?`	Size of the source when known
`Warnings`	`IReadOnlyList<PdfWarning>`	Non-fatal remarks about the document

Two computed properties save a loop: `FirstPageSize` is the first page's size or `null` for an empty document, and `HasUniformPageSize` is `true` when every page has the same size — convenient for deciding whether a merge will produce a uniform document.

`PdfPageInfo` describes one page:

Property	Type	Meaning
`Number`	`int`	The one-based page number
`Size`	`PdfPageSize`	The size as stored, before rotation
`Rotation`	`PageRotation`	How a viewer rotates the page before displaying it
`HasContent`	`bool`	Whether the page has a content stream

Property	Type	Meaning
`AnnotationCount`	`int`	Number of annotations on the page
`DisplaySize`	`PdfPageSize`	`Size` with the rotation applied

`PdfPageSize` is a value object with `WidthPoints` and `HeightPoints` as the source of truth, plus the millimetre and inch conversions, `IsLandscape`, `IsPortrait` and the predefined `A4`, `A5`, `A3`, `Letter` and `Legal` sizes. Use `DisplaySize`, not `Size`, when you care about what a reader sees: a landscape page is frequently stored as a portrait page plus a 90-degree rotation.

```C#
Console.WriteLine($"{info.SourceName}: PDF {info.Version}, {info.PageCount} page(s)");

if (info.FirstPageSize is { } size)
{
    Console.WriteLine($"First page: {size.WidthMillimeters:0} x {size.HeightMillimeters:0} mm");
}

if (!info.HasUniformPageSize)
{
    var sizes = info.Pages
        .Select(page => page.DisplaySize)
        .Distinct()
        .Select(size => size.ToString());

    Console.WriteLine($"Mixed page sizes: {string.Join(", ", sizes)}");
}

Console.WriteLine(info.HasOutlines
    ? $"{info.OutlineItemCount} bookmark(s)"
    : "No bookmarks");
```

What inspection is not

Inspection reads structure, not content. It does not extract text, render pages, count words or check that a font is embedded. `HasContent` says only that a content stream exists, not that it draws anything visible. `AnnotationCount` counts annotations, not the form fields they belong to.

It is also not free. The inspector parses the document, including its page tree, so the cost is close to reading the same document for a merge. Calling it on every request to display a page count doubles the work; if you need both, merge and read the count from `MergeResult.PageCount` instead, or cache the `PdfDocumentInfo` returned by one inspection.

Encrypted and damaged documents

Inspection has a different failure contract from merging, and the difference is deliberate.

- An **encrypted document whose encryption stayed closed** does not throw. You get a `PdfDocumentInfo` with `IsEncrypted = true`, `IsReadable = false`, `RequiresPassword` telling you whether a password would help, and a warning explaining what happened. `PageCount` is zero and `Pages` is empty, but `SourceName`, `SecurityHandlerRevision` and `Warnings` are populated.
- A **malformed document that is not a readable PDF** throws `PdfSourceFormatException`. There is no partial result to return, because the object structure could not be established.
- **A document that is readable but damaged** does not throw. The reader repairs what it can — rebuilds a broken cross-reference table, skips an unreadable page-tree branch, assumes A4 for a page without a media box — and reports each repair as a warning.

```C#
var info = await inspector.InspectAsync(new PdfFileSource(path), password);

if (!info.IsReadable)
{
    Console.WriteLine(info.RequiresPassword
        ? "The document is encrypted; supply the user or owner password."
        : "The document is encrypted with a security handler this library does not implement.");
}
```

`RequiresPassword` is about the *user* password. A document that opens without a password but is restricted against editing has `IsEncrypted = true`, `IsReadable = true` and `RequiresPassword = false`; its restrictions do not block inspection or merging. `PdfDocumentInfo` does not expose the permission flags themselves — inspect the encryption dictionary with another tool if you need to display them.

Warnings

`PdfWarning` pairs a stable `Kind` with a human-readable `Message`. The kind is the contract; the text may be reworded in any release, so match on `Kind` and log `Message`. The set of kinds is open, which means a `switch` over `PdfWarningKind` must keep a `default` arm.

The kinds fall into four groups:

Group	Examples	What it means
Structure repaired	`CrossReferenceRebuilt`, `CrossReferenceEntryMismatch`, `ObjectStreamTruncated`	The reader recovered; the document may be damaged
Page tree	`PageTreeMissing`, `PageTreeCycle`, `PageSizeAssumed`	Pages were skipped, cut off, or described with a default size
Encryption	`EncryptionOpened`, `EncryptionHandlerUnsupported`, `CryptFilterUndefined`	The encryption was understood, opened, or could not be used
Merge decisions	`NamedDestinationsDropped`, `OutlinesDroppedByPolicy`, `BookmarksTruncated`	Content was intentionally not carried over

`EncryptionOpened` is a warning even though it describes success: it tells a caller that the document was protected and is not simply unprotected. `OutputVerified` is likewise a success signal, reported only when `PdfEngineOptions.VerifyOutput` is enabled (Chapter 10).

Warnings from a source are folded into `MergeResult.Warnings`, so a caller who only looks at the merge result still learns that a source had to be repaired. Identical warnings are not repeated; because the message names the source, two damaged sources normally produce two warnings rather than one.

A practical handling policy:

```C#
foreach (var warning in result.Warnings)
{
    logger.LogWarning("{Kind}: {Message}", warning.Kind, warning.Message);
}
```

Treat structural warnings as reasons to keep the source but investigate it, and treat merge-decision warnings as expected when the plan deliberately drops something. Do not fail a merge because warnings exist: if a condition was fatal, the merge would have thrown.

Using inspection as a preflight

The inspector is most useful immediately before a merge:

1. Inspect each source with its password.
2. Reject documents that are unreadable, and ask for a password when `RequiresPassword` is true.
3. Resolve page selections against `PageCount` so a bad range is a validation message rather than an exception halfway through writing (Chapter 5).
4. Warn the user about repairs, encryption and unusual page sizes.
5. Merge only the sources that passed.

This mirrors the order the merger itself uses — open, read structure, resolve selections, write — but gives you a place to interject a message, a prompt or a policy decision. It does not make the merge safer in the sense of catching every error: a file can change between inspection and merge, and the merger validates again. The gain is a better conversation with the user, not a substitute for handling `PdfException` (Chapter 9).

9. Errors, Exceptions and Warnings

SharpCorePDF separates three kinds of bad news:

- **Programmer errors** — a null argument, a stream that cannot be written, invalid page notation — throw the standard .NET exceptions (`ArgumentNullException` , `ArgumentException` , `FormatException`). They indicate a bug in the calling code, not a problem with a document.
- **Document problems** — a malformed source, a missing password, an unsupported compression filter, an impossible plan — throw a subclass of `PdfException` .
- **Non-fatal remarks** — a repaired cross-reference table, a bookmark that had to be skipped, a feature deliberately dropped by policy — are returned in `Warnings` and never throw.

Everything in the third group is reported on `PdfDocumentInfo.Warnings` when inspecting and on `MergeResult.Warnings` after a merge. Treating a warning as an error is a policy decision, not something the library forces on you.

The exception family

`PdfException` is abstract and is the base of every PDF-related error. Catching it catches all of them.

Exception	Meaning	Useful members
`MergePlanValidationException`	A plan invariant is broken, or a page selection does not fit its source	`Errors`
`PdfSourceFormatException`	A source is not a readable PDF, or its structure cannot be trusted	`SourceName` , `Detail`
`PdfEncryptionException`	A source is encrypted and could not be opened	`SourceName` , `RequiresPassword`
`PdfUnsupportedFeatureException`	A source uses a PDF feature the engine deliberately does not implement	`SourceName` , `Feature`

`MergePlanValidationException` is unusual in carrying more than one message. `Errors` lists every problem found in one pass, and `Message` joins them. Show the list if you are reporting to a developer; show the first item if you are reporting to a user.

`PdfSourceFormatException` and `PdfEncryptionException` name the offending source, which matters when a plan has ten sources. `PdfUnsupportedFeatureException` names the feature — for example an unsupported compression filter for an object stream — so the message can say what the tool would need to support rather than only that the file failed.

What throws when

The sequence inside `MergeAsync` explains where each exception comes from:

1. `plan.EnsureValid()` runs first. A plan with no sources, a primary source that is not in the plan, a blank display name, an undefined policy value or invalid encryption settings throws `MergePlanValidationException` before any file is opened.
2. Each source is opened and read. A missing file throws `IOException` or `UnauthorizedAccessException` from the content source; malformed content throws `PdfSourceFormatException` ; a source that stays encrypted throws `PdfEncryptionException` .
3. Each page selection is resolved against the real page count. A range that starts past the last page, or a document with no pages, throws `MergePlanValidationException` .
4. The engine writes pages. Unsupported filters and malformed objects discovered here still surface as `PdfException` subclasses, not raw index errors.
5. Cancellation is observed between sources and during writing and surfaces as `OperationCanceledException` , the standard .NET contract.

Because steps 1 and 3 happen before the engine writes, a bad plan or a bad page range does not produce a partial PDF in the destination stream. It does not protect an existing file, however: `MergeToFileAsync` creates and truncates the destination before it calls the merger. The section on writing files safely later in this chapter shows the pattern that avoids this.

Handling exceptions

Catch specific types from the most specific to the least, so each gets the response it deserves:

```C#
try
{
    plan.EnsureValid();
    var result = await merger.MergeToFileAsync(plan, destination, cancellationToken);
    Console.WriteLine(result);
}
catch (MergePlanValidationException exception)
{
    // The caller can fix this: bad range, empty plan, missing primary source.
    foreach (var error in exception.Errors)
    {
        Console.WriteLine($"plan: {error}");
    }
}
catch (PdfEncryptionException exception) when (exception.RequiresPassword)
{
    // Ask for a password for exception.SourceName and retry.
    Console.WriteLine($"'{exception.SourceName}' needs a password.");
}
catch (PdfEncryptionException exception)
{
    // Unsupported security handler: no password will help.
    Console.WriteLine(exception.Message);
}
catch (PdfSourceFormatException exception)
{
    Console.WriteLine($"'{exception.SourceName}' is damaged: {exception.Detail}");
}
catch (PdfUnsupportedFeatureException exception)
{
    Console.WriteLine($"'{exception.SourceName}' uses {exception.Feature}: {exception.Message}");
}
catch (PdfException exception)
{
    // Any PDF-related failure that the cases above did not name.
    Console.WriteLine(exception.Message);
}
catch (OperationCanceledException)
{
    Console.WriteLine("The merge was cancelled.");
}
catch (IOException exception)
{
    Console.WriteLine($"The file system refused the merge: {exception.Message}");
}
```

Three details are easy to get wrong:

- **`OperationCanceledException` is not a `PdfException`.** It derives from `SystemException` directly, so a `catch (PdfException)` will not see a cancellation. Catch it separately, and do not log it as a failure.
- **`IOException` is not a `PdfException` either.** A missing input file, a read-only destination or a full disk is a file-system problem. It can occur at any step, including while a source is read.
- **Order matters.** `PdfEncryptionException` and `PdfSourceFormatException` both derive from `PdfException`; catching the base first would hide the members that make them actionable.

`ArgumentException`, `ArgumentNullException` and `FormatException` are deliberately not caught in this example. They mean the code built the plan incorrectly — a null stream, an unwritable destination, a malformed page string — and the fix belongs in the code, not in a retry loop. Validate page text with `PageSelection.TryParse` before building the plan, so a typo becomes a validation message rather than an exception in the middle of a request.

Writing files safely

`MergeToFileAsync` is convenient and is the right call for a scratch file, but it creates the destination with `FileMode.Create` before the merge begins. If the merge then fails or is cancelled, the destination is left empty or partially written. Two consequences:

1. **Never merge a file onto itself.** If the destination is also a source path, the source is truncated before it is read. The merge fails and the original is gone.
2. **Do not treat a returned file as committed until the call returns successfully.** A watcher that processes the directory may see a partial file while the merge is still running.

For anything that matters, write to a temporary file in the same directory and move it into place only after success:

```C#
var fullPath = Path.GetFullPath(destination);
var directory = Path.GetDirectoryName(fullPath)!;
Directory.CreateDirectory(directory);

var temporary = Path.Combine(directory, $".{Path.GetFileName(fullPath)}.{Guid.NewGuid():N}.tmp");

try
{
    await merger.MergeToFileAsync(plan, temporary, cancellationToken);
    File.Move(temporary, fullPath, overwrite: true);
}
finally
{
    if (File.Exists(temporary))
    {
        File.Delete(temporary);
    }
}
```

The temporary file lives in the same directory, so the move stays on one volume, where it is atomic on Windows and on the common Unix file systems. The `finally` block removes the temporary file on failure and on cancellation; after a successful move the file is already gone. The destination is either the previous file or the complete new one, never a half-written document.

`MergeToBytesAsync` has the same all-or-nothing property for a different reason: the buffer is local to the call, so a failure returns no bytes and leaves nothing behind. It is the simplest safe choice when the result fits in memory.

Warnings versus errors

A warning never changes the outcome: the merge completed and the result is usable. The distinction is worth preserving in your own API, because collapsing warnings into errors makes the library look less reliable than it is, and ignoring them hides real repairs.

A reasonable policy:

Warning group	Suggested response
Structure repaired (`CrossReferenceRebuilt`, `ObjectStreamTruncated`)	Log; keep the source; investigate if it recurs
Page tree (`PageTreeMissing`, `PageSizeAssumed`)	Log at warning level; compare the page count with expectations
Encryption (`EncryptionOpened`)	Log at information level; the password worked

Warning group	Suggested response
Merge decisions (`OutlinesDroppedByPolicy` , `NamedDestinationsDropped`)	Log at debug level; expected when the plan drops features
`OutputVerified`	Log at information level; verification was requested and passed

If your product has a "strict" mode, implement it in your code by inspecting `result.Warnings` and deciding which kinds are unacceptable — do not expect the merger to throw for them. Chapter 8 lists the kinds and their meanings.

Reproducing a failure

When a merge fails, the report should contain the plan and the exception, not only the message:

```
C#

catch (PdfException exception)
{
    logger.LogError(exception, "Merge failed. Plan: {Plan}", plan.ToString());
}
```

`MergePlan.ToString()` summarises the source count, target version and policies without printing page content. `PdfDocumentInfo.ToString()` and `MergeResult.ToString()` do the same for a document and a completed merge. None of them includes passwords: `PdfSource.ToString()` prints the display name and page selection only.

Do not log the plan *object* with structured logging, though. The properties are public, and `MergeOutputOptions` is a record whose generated `ToString()` would include the encryption options and their passwords. Log `plan.ToString()` or the individual non-secret fields.

10. Dependency Injection

The `SharpCorePDF.DependencyInjection` package registers the merger, the inspector and the engine behind their ports. It is a thin layer: one call, no auto-discovery, no background services and no logging of its own. The `SharpCorePDF` meta-package already depends on it, so the types are available after the installation described in Chapter 3.

One call

```C#
using SharpCorePDF.DependencyInjection;

var builder = WebApplication.CreateBuilder(args);

builder.Services.AddSharpCorePDF();

var app = builder.Build();
```

That registers the services listed below as singletons. They are stateless; a merge creates its own context, so concurrent requests can use the same `IPdfMerger` instance without coordination.

Service	Lifetime	Notes
`PdfEngineOptions`	Singleton	`PdfEngineOptions.Default` unless replaced
`PdfDocumentReader`	Singleton	Concrete reader; registered for advanced use
`IPdfDocumentReader`	Singleton	Resolves to the same `PdfDocumentReader`
`PdfMergeEngine`	Singleton	Concrete engine
`IPdfMergeEngine`	Singleton	Resolves to the same `PdfMergeEngine`
`IPdfMerger`	Singleton	`PdfMergerService`, the use case to depend on
`IPdfInspector`	Singleton	`PdfInspectorService`

Application code should depend on `IPdfMerger` and `IPdfInspector`, not on the concrete services. The concrete registrations exist so a test can replace one layer without replacing the whole stack, and so the engine can be exercised directly.

```C#
app.MapPost("/merge", async (IPdfMerger merger, IFormFileCollection files) =>
{
    var plan = MergePlan.Create();

    foreach (var file in files)
    {
        await using var stream = file.OpenReadStream();
        using var buffer = new MemoryStream();
        await stream.CopyToAsync(buffer);
        plan = plan.AddBytes(buffer.ToArray(), file.FileName);
    }

    var bytes = await merger.MergeToBytesAsync(plan);
    return Results.File(bytes, "application/pdf", "merged.pdf");
});
```

Each `AddBytes` call creates a source with a fresh identity, so uploading the same file twice contributes its pages twice. The bytes are buffered before the merge because the merger reads its sources later; an `IFormFile` stream is not guaranteed to stay open until then. Chapter 4 covers the other factories.

Without the container

`AddSharpCorePDF()` only saves typing. The reader, the engine and the two services have public constructors, so the same graph can be built by hand:

```C#
using SharpCorePDF.Application.Services;
using SharpCorePDF.Domain.Ports;
using SharpCorePDF.Infrastructure;

var options = PdfEngineOptions.Default;
var reader = new PdfDocumentReader(options);

IPdfMerger merger = new PdfMergerService(reader, new PdfMergeEngine(options));
IPdfInspector inspector = new PdfInspectorService(reader);
```

The two routes are equivalent: the container builds exactly this graph and registers it as singletons. The `no-container` example in the repository runs the same plan through both and compares the bytes.

Three ways to configure the engine

The extension has three overloads; the configuration overload takes an optional section name, so these four shapes are possible:

```C#
services.AddSharpCorePDF();                                    // defaults
services.AddSharpCorePDF(PdfEngineOptions.Default with { ... }); // explicit options
services.AddSharpCorePDF(configuration);                       // the "SharpCorePDF" section
services.AddSharpCorePDF(configuration, "My:Pdf:Section");     // a section of your choosing
```

- **No argument** registers `PdfEngineOptions.Default` through `TryAddSingleton`, so an options instance you registered earlier is left in place.
- **An options instance** is registered with `Replace`, so it wins over an earlier registration.
- **A configuration** binds the named section and registers the bound value. The default section name is the `DefaultConfigurationSection` constant, `"SharpCorePDF"`.

The configuration overload does not fail when the section is missing: binding an absent section produces the default values. That is convenient in development and dangerous in production, where a typo in the section name silently disables a limit you thought you had set. Validate explicitly:

```C#
var engineOptions = builder.Configuration
    .GetSection(SharpCorePDFServiceCollectionExtensions.DefaultConfigurationSection)
    .Get<PdfEngineOptions>() ?? PdfEngineOptions.Default;

var problems = engineOptions.Validate();

if (problems.Count > 0)
{
    throw new InvalidOperationException(
        $"SharpCorePDF configuration is invalid: {string.Join(" ", problems)}");
}

builder.Services.AddSharpCorePDF(engineOptions);
```

`Validate()` checks the numeric limits and returns every problem it finds instead of throwing. It does not read any document, so it cannot catch a page selection that reaches past the last page; that surfaces when the plan runs (Chapter 9).

Engine options

`PdfEngineOptions` is an immutable record. Configure it with `with` expressions, and remember that the defaults are deliberately permissive:

Option	Default	Effect
`MaxSourceBytes`	512 MiB	Largest source read into memory; larger sources are rejected
`MaxNestingDepth`	64	Maximum dictionary/array nesting and page-tree depth
`MaxPagesPerSource`	100 000	Declared page limit; see the note below
`ReconstructCrossReferenceOnFailure`	`true`	Rebuild a damaged or missing cross-reference table by scanning
`VerifyOutput`	`false`	Re-read the written document and check its page count
`WriteBinaryMarker`	`true`	Write the binary comment after the `%PDF` header

The source limit is enforced by the reader, and exceeding it throws `PdfSourceFormatException` with a detail that names the limit. It is not a streaming cap: the whole source is held in memory while it is processed, so the practical ceiling is your process's memory budget, not the number on the option.

`MaxNestingDepth` has two effects. It bounds how deep an object graph may be parsed, and it bounds the page-tree walk; a page tree that is cut off contributes a `PageTreeTooDeep` warning rather than failing the merge.

`ReconstructCrossReferenceOnFailure` is the difference between "repair what you can" and "reject what is not well formed". Leaving it on produces a `CrossReferenceRebuilt` warning when repair happened. Turning it off is appropriate when a damaged file should be treated as an error, for example in a regulatory archive.

`VerifyOutput` is a development and test switch. It captures the complete output in memory, reads it back and compares the page count with what was written, then adds an `OutputVerified` warning on success. That doubles the work and holds the whole result in memory, which defeats the streaming design for large documents. Use it in integration tests, not in production.

`WriteBinaryMarker` should stay on. Some transfer tools inspect the first bytes to decide whether a file is binary; the marker costs a few bytes and prevents a class of corruption that is otherwise very hard to diagnose.

> `MaxPagesPerSource` is part of the options record but this version does not enforce it. Do not rely on it as a security boundary. For untrusted input, set `MaxSourceBytes`, inspect the document with `IPdfInspector` and reject it if `PageCount` exceeds your own limit, before building a plan.

Configuration file

With the configuration overload, the section names map directly to the property names:

JSON

```
{
  "SharpCorePDF": {
    "MaxSourceBytes": 268435456,
    "MaxNestingDepth": 64,
    "MaxPagesPerSource": 50000,
    "ReconstructCrossReferenceOnFailure": true,
    "VerifyOutput": false,
    "WriteBinaryMarker": true
  }
}
```

Environment variables and command-line arguments follow the normal configuration rules, for example `SharpCorePDF__MaxSourceBytes`. Because the section is bound to an immutable record, a property that is missing keeps its default; a property with a value of the wrong type causes a binding exception at start-up, which is one of the few failures you want to be loud.

Logging and diagnostics

The library has no `ILogger` dependency and writes nothing to the console. That is deliberate: a class library that logs through a static logger cannot be tested cleanly and cannot respect a host's configuration. Everything you need to log is in the return values:

- `MergeResult` carries `PageCount`, `SourceCount`, `OutputBytes`, `OutputVersion`, `Duration`, `PrimarySourceId`, per-source summaries and `Warnings`.
- `PdfDocumentInfo` carries the same kind of facts for inspection, plus `IsReadable`, `IsEncrypted` and `RequiresPassword`.
- Exceptions carry `SourceName`, and `MergePlanValidationException` carries `Errors`.

Log them at the boundary of your application, where you know the request and the user. Chapter 9 shows the exception-handling shape, and Chapter 8 explains the warning kinds.

The producer string

When `MergeOutputOptions.Producer` is not set, the writer stamps the document with the product name and the assembly version of the SharpCorePDF domain assembly, for example `SharpCorePDF 1.0.0`. Override it with your own product name if the PDF is produced by your application rather than offered as a library feature:

```C#
var plan = MergePlan.Create()
    .AddFile("input.pdf")
    .WithOutput(options => options with { Producer = "Contoso Report Server 4.2" });
```

There is no separate "application name" setting in `PdfEngineOptions`; the producer belongs to the output document, not to the engine, and therefore lives on `MergeOutputOptions` (Chapter 6).

Testing with the container

For integration tests, register the real stack and turn verification on:

```C#
var services = new ServiceCollection();
services.AddSharpCorePDF(new PdfEngineOptions { VerifyOutput = true });
services.AddSingleton<MyReportService>();

await using var provider = services.BuildServiceProvider();
var report = provider.GetRequiredService<MyReportService>();

var bytes = await report.BuildAsync();
Assert.True(bytes.Length > 0);
```

An `InvalidOperationException` from the merge in this configuration means the engine wrote a document it could not read back — a defect worth reporting, not a problem with the input. Ordinary input problems still appear as `PdfException` subclasses.

To test the layer above, replace `IPdfMerger` or `IPdfInspector` with a fake after the registration. Because the concrete services are registered separately, replacing only the interface leaves the engine available for tests that need it.

11. Architecture

SharpCorePDF is built as four layers around a pure domain model. This chapter explains the shape and the reasons behind it, so that you can judge where a change belongs and what a failure means.

The layers

Project	Contains	Depends on
`SharpCorePDF.Domain`	`MergePlan`, `PdfSource`, value objects, policies, read models, warnings, exceptions, `IPdfContentSource`	Nothing but the BCL
`SharpCorePDF.Application`	`PdfMergerService`, `PdfInspectorService`, `IPdfDocumentReader`, `IPdfMergeEngine`, extension methods	Domain
`SharpCorePDF.Infrastructure`	The PDF reader, writer, object model, filters, copier, engine and content sources	Domain, Application
`SharpCorePDF.DependencyInjection`	`AddSharpCorePDF()` and the service registrations	Domain, Application, Infrastructure
`SharpCorePDF` (meta-package)	No assembly; depends on the four above	—

Dependencies point inward:

```
TEXT
SharpCorePDF (meta)
        │
        ├── DependencyInjection ──┬── Infrastructure ──┐
        │                         ├── Application ─────┤
        │                         └── Domain ◄─────────┘
        └── (transitively, all four)
```

The domain does not know what a file is. It defines plans, page selections, output options and results, but the only way it reaches the outside world is `IPdfContentSource`, which is a stream factory with a display name. That is why a plan can be built, validated and tested without touching the disk.

The application layer orchestrates but does not parse PDF syntax. It validates a plan, opens each source, asks `IPdfDocumentReader` what is inside, resolves the page selections against the real page counts, and hands a `MergeEngineRequest` to `IPdfMergeEngine`. It also owns the lifetime of every stream it opens, so a failure in the middle does not leak a handle.

The infrastructure layer is the PDF engine. It reads bytes, parses objects, decrypts, walks page trees, copies object graphs, writes a cross-reference table and encrypts the result. It has no opinion about where a document came from or which pages the caller wanted; the request states both.

The dependency injection project is the composition root. It knows the concrete classes, and nothing else does.

The life of a merge

A call to `IPdfMerger.MergeAsync` passes through these stages:

1. **Argument checks.** A null plan or destination is a programmer error.
2. **Plan validation.** `EnsureValid()` checks the invariants that do not need a document: at least one source, unique identities, a primary source that belongs to the plan, defined enum values, usable encryption settings.
3. **Opening.** Each `IPdfContentSource` is asked for a readable stream. The streams stay open until the merge ends.
4. **Inspection.** `IPdfDocumentReader` reads each source, opens its encryption if a password was given, walks its page tree and reports `PdfDocumentInfo`. The application layer turns an unreadable encrypted source into `PdfEncryptionException` here, while it still knows which source it was.

5. **Resolution.** `PageSelection.Resolve` maps the one-based ranges onto real page numbers, collapses duplicates and preserves order; the engine skips a closed range's tail past the last page.
6. **Writing.** The engine receives the resolved sources and the output options. It copies the objects that the selected pages and retained features reach, renumbers them, builds one page tree, merges catalog-level features according to the policies, and writes the result to the destination.
7. **Release.** The application disposes every source stream in a `finally`, including on cancellation and on failure.

The engine never mutates the plan or the request. Everything it needs while running lives in `PdfMergeContext`, a per-merge object; that is what makes the registered services safe to share.

Why the plan is immutable

`MergePlan` is an immutable record. `AddSource`, `WithOutput` and the rest return new instances, and a source identity is created once and preserved by every operation on `PdfSource`. Three consequences are deliberate:

- A plan can be validated and then executed without a window in which another thread changes it.
- `MergePlanValidationException.Errors` describes a specific plan value, not a transient state.
- A plan is safe to log, cache or pass across layers. Its `ToString()` is stable and excludes passwords.

The cost is allocation on every edit. Building a plan of a hundred sources allocates a hundred plans; that is negligible next to reading the sources, and it buys predictable behaviour.

What the engine writes

The result is a new PDF, not an incremental update of a source. In particular:

- The merged document uses a classic cross-reference table, even when a source used cross-reference streams or object streams. Objects stored in object streams are written as ordinary objects.
- Only reachable objects are copied. A source's unused fonts, unused images or long-deleted revisions do not travel into the result.
- The object graph of each source is copied in two phases — reserve, then write — so cycles such as a form field pointing at its widget and back are handled without recursion loops.
- A single page tree node lists the selected pages in merge order. Page labels, outlines, annotations, form fields and attachments are handled by the output policies described in Chapter 6.
- The document is written as it is produced; there is no temporary full copy in the engine unless `VerifyOutput` is enabled, which captures the whole output so it can be re-read and checked.

The memory model

Two facts determine the memory profile, and both are worth knowing before processing large files:

1. **Each source is read completely into memory.** `PdfByteReader` reads a stream into a pooled buffer, and parsing resolves objects from those bytes on demand. `MaxSourceBytes` is the guard.
2. **Every source is held until the merge finishes.** The application inspects each source once, then the engine reads it again, and the engine keeps a reader per source while it copies objects from all of them. Peak memory is therefore roughly the sum of all source sizes, plus the destination's own buffering.

The destination is written in a streaming fashion, so a large result does not need a second full copy in memory. Verification is the exception: `VerifyOutput` deliberately captures the whole output.

For very large jobs, split the work into separate output documents, or merge subsets to temporary files and keep the intermediates on disk rather than in byte arrays. Staging reduces the work of each step, but the final merge still holds every intermediate it is given; it cannot lower the peak below the total size of all sources for one output document.

Boundaries and extension points

The ports are the seams:

Port	Implemented by	Replace it when
`IPdfContentSource`	`PdfFileSource`, byte and stream sources	Documents come from a database, blob store or network
`IPdfDocumentReader`	`PdfDocumentReader`	You need a different inspection strategy in tests
`IPdfMergeEngine`	`PdfMergeEngine`	You want to stub the writer while testing orchestration
`IPdfMerger`	`PdfMergerService`	You want to test a caller without any PDF work
`IPdfInspector`	`PdfInspectorService`	Same, for inspection

A content source is small enough to write for any backing store. Each call must return an independent, readable stream positioned at the start, and `DisplayName` should identify the document in errors:

```C#
using SharpCorePDF.Domain.Ports;

public sealed class StreamFactorySource(
    string displayName,
    Func<CancellationToken, ValueTask<Stream>> open) : IPdfContentSource
{
    public string DisplayName { get; } = displayName;

    public long? Length => null;

    public ValueTask<Stream> OpenReadAsync(CancellationToken cancellationToken = default) =>
        open(cancellationToken);
}
```

```C#
var source = PdfSource.Create(new StreamFactorySource(
    "report.pdf",
    cancellationToken => storage.OpenReadAsync("report.pdf", cancellationToken)));
```

The merger takes ownership of the returned stream and disposes it, so the factory must return a fresh stream on every call. If the backing store can report a length cheaply, return it from `Length`; the current reader buffers from the stream's own `Length` when it is seekable, so the property is a declaration of what you know rather than something the engine relies on today.

What is *not* pluggable is equally deliberate: PDF filters, encryption handlers and the object model live inside the infrastructure layer. Adding a filter means changing the engine, not registering a service. The supported feature set is a compatibility promise, and a plugin point would make it impossible to say what the library can read.

Concurrency

Every registered service is a singleton and stateless. Concurrent merges share the services but not the data: each call has its own request, context, readers and writer. Two rules follow for callers:

- **Do not share a destination stream between concurrent merges.** The stream itself is not synchronised, and interleaved PDF output is corrupt by construction.
- **Do share plans.** An immutable plan can be executed repeatedly and concurrently; the sources open a fresh stream per merge.

The content source is the only collaborator a caller supplies, and the contract requires `OpenReadAsync` to be safe to call more than once. A source backed by a non-seekable one-shot network stream cannot honour that contract; buffer it first, as the upload example in Chapter 10 does.

12. Cookbook

Short, complete recipes for the shapes that come up most often. Each one assumes `IPdfMerger` and `IPdfInspector` are available, as described in Chapter 10, and that the destination path is not one of the input files.

Merge every PDF in a folder, in a defined order

`EnumerateFiles` returns files in an unspecified order on most platforms. Sort explicitly, and sort numerically if your files are numbered without zero padding.

```C#
var files = Directory
    .EnumerateFiles(folder, "*.pdf", SearchOption.TopDirectoryOnly)
    .OrderBy(path => Path.GetFileName(path), StringComparer.OrdinalIgnoreCase)
    .ToArray();

if (files.Length == 0)
{
    throw new InvalidOperationException($"No PDF files found in '{folder}'.");
}

var plan = MergePlan.Create();

foreach (var file in files)
{
    plan = plan.AddFile(file);
}

await merger.MergeToFileAsync(plan, Path.Combine(folder, "merged.pdf"));
```

`merged.pdf` itself matches `*.pdf`; on a second run the plan would include the previous result and the destination would truncate it before the merge reads it. Filter the output name out, write outside the folder, or use the temporary-file pattern from Chapter 9.

For `report-2.pdf` before `report-10.pdf`, the default comparator is not enough: it compares text, so `report-10.pdf` sorts before `report-2.pdf`. Zero-pad the names, or sort with a natural-order comparer that understands embedded numbers.

Ask for a password only when one is needed

Inspection never throws for an encrypted document it cannot open; it reports it.

```C#
var content = new PdfFileSource(path);
var info = await inspector.InspectAsync(content, cancellationToken: cancellationToken);

string? password = null;

if (!info.IsReadable)
{
    if (!info.RequiresPassword)
    {
        throw new InvalidOperationException(
            $"'{info.Metadata.Title ?? Path.GetFileName(path)}' cannot be opened, and a password would not
help.");
    }

    password = PromptForPassword(path);

    if (password is null)
    {
        return;
```

```
    }

    info = await inspector.InspectAsync(content, password, cancellationToken);

    if (!info.IsReadable)
    {
        throw new InvalidOperationException("The password did not open the document.");
    }
}

var plan = MergePlan.Create().AddFile(path, password: password);
await merger.MergeToFileAsync(plan, destination, cancellationToken);

static string? PromptForPassword(string path)
{
    Console.Write($"Password for '{Path.GetFileName(path)}': ");
    return Console.ReadLine();
}
```

Two details matter. A document that is only restricted against editing opens with no password at all, so `password` stays `null` and the merge succeeds — asking for a password there would be confusing. And `InspectAsync` must be repeated with the password if you want to use the page count for a range; `info.PageCount` from the unreadable result is not trustworthy.

`Console.ReadLine` echoes the password. In a real client, use the password control of your UI or a masked console helper; the branching logic stays the same.

Extract a page range

Extraction is a merge of one source with a selection:

```C#
var plan = MergePlan.Create()
    .AddFile("manual.pdf", pages: "3-7,12")
    .WithOutput(options => options with
    {
        OutlinePolicy = OutlinePolicy.Preserve,
        MetadataPolicy = MetadataPolicy.FromPrimarySource,
    });

await merger.MergeToFileAsync(plan, "excerpt.pdf");
```

The result contains six pages in the order written: 3, 4, 5, 6, 7, 12. With `OutlinePolicy.Preserve` the bookmarks of the source are carried over and renumbered onto the pages of the excerpt; the policy is the default. Page labels are dropped by default because a label such as "iv" no longer describes the same position in the new document. Set `PageLabelPolicy = PageLabelPolicy.FromPrimarySource` only when the copied labels still make sense from the first output page, as they do when the selection starts at page 1.

Encrypt the result

```C#
var options = new PdfEncryptionOptions
{
    Algorithm = PdfEncryptionAlgorithm.AesWith256BitKey,
    UserPassword = "reader-secret",
    OwnerPassword = "owner-secret",
    Permissions = PdfPermissions.PrintOnly,
};

var plan = MergePlan.Create()
    .AddFile("internal.pdf")
    .WithOutput(output => output with
    {
        TargetVersion = PdfVersion.V1_7,
        Encryption = options,
    });

await merger.MergeToFileAsync(plan, "internal-protected.pdf");
```

AES-256 requires PDF 1.7, which is the default target, so the explicit version is only there to show the dependency. The reader who supplies `reader-secret` may print but not copy text or modify the document; the owner password unlocks every permission. If you omit `OwnerPassword`, the user password acts as the owner password too — convenient, but then anyone who can open the document can also remove the restrictions.

Merge uploads in an ASP.NET Core endpoint

`IFormFile` gives you a stream; `AddStream` accepts one and reads it exactly once at merge time. Merging to bytes keeps the response simple because a byte array has a known length:

```C#
[HttpPost("merge")]
[RequestSizeLimit(100_000_000)]
public async Task<IActionResult> MergeAsync(
    [FromForm] List<IFormFile> files,
    CancellationToken cancellationToken)
{
    if (files.Count == 0)
    {
        return BadRequest("Upload at least one PDF.");
    }

    var plan = MergePlan.Create();

    foreach (var file in files)
    {
        plan = plan.AddStream(file.OpenReadStream(), file.FileName);
    }

    var merged = await merger.MergeToBytesAsync(plan, cancellationToken);

    return File(merged, "application/pdf", "merged.pdf");
}
```

The streams belong to the request and are read before the action returns, so no upload outlives the request. A total request limit is not optional: each source is buffered in full, so an unlimited upload is an unlimited allocation. If the result may be large, write it to a temporary file and return a `PhysicalFile` or `FileStreamResult` instead of holding a second copy in memory.

`MergeToBytesAsync` is all-or-nothing: an exception leaves no partial response body to send, because the body is only produced after the merge succeeded. Streaming the merge directly to `Response.Body` would be cheaper but sends a truncated document when a later source fails.

Report what was written

The result is meant to be logged as-is. Do not parse the text of a warning for control flow; branch on its `Kind`, which is the stable part of the contract.

```C#
var result = await merger.MergeToFileAsync(plan, destination, cancellationToken);

logger.LogInformation(
    "Merged {SourceCount} source(s) into {PageCount} page(s), {OutputBytes} bytes, PDF {Version} in {Duration}
ms.",
    result.SourceCount,
    result.PageCount,
    result.OutputBytes,
    result.OutputVersion,
    result.Duration.TotalMilliseconds);

foreach (var source in result.Sources.Where(source => !source.IsComplete))
{
    logger.LogInformation(
        "{SourceName}: {PagesContributed} of {SourcePageCount} pages selected.",
        source.SourceName,
        source.PagesContributed,
        source.SourcePageCount);
}

foreach (var warning in result.Warnings)
{
    logger.LogWarning("{WarningKind}: {WarningMessage}", warning.Kind, warning.Message);
}
```

`SourceCount` counts the sources of the plan, not distinct files: adding the same file twice produces two entries. `OutputBytes` is the size of the document the engine wrote; for `MergeToFileAsync` the file has already been flushed when the call returns, so it matches the file length.

How many warnings are too many is a policy decision. A production service usually logs known-benign kinds at information level and the rest at warning level, and keeps the warning kinds in its metrics so a rise in repaired documents is visible before users report it.

Override metadata and fix the modification date

`MetadataOverride` replaces only the fields you set. Everything else still follows `MetadataPolicy`, which is `FromPrimarySource` by default:

```C#
var created = new DateTimeOffset(2026, 1, 1, 0, 0, 0, TimeSpan.Zero);

var plan = MergePlan.Create()
    .AddFile("cover.pdf")
    .AddFile("body.pdf")
    .WithOutput(options => options with
    {
        MetadataOverride = new PdfMetadata
        {
            Title = "Annual Report 2026",
            Author = "Contoso",
            Subject = "Results for the financial year",
            Keywords = "annual report, 2026, results",
        },
        Producer = "Contoso Report Server 4.2",
        ModificationDate = created,
    });

await merger.MergeToFileAsync(plan, "annual-report.pdf");
```

The primary source is the first source unless `WithPrimarySource` says otherwise, so the title you did not override, the custom properties and the creation date come from `cover.pdf`. Viewer preferences, page layout and page mode are also taken from the primary source, but they are catalog entries rather than document information and are not affected by `MetadataPolicy`. To drop the metadata fields you did not override, set `MetadataPolicy = MetadataPolicy.None`; the policy starts from an empty information dictionary and the override is still applied on top. Chapter 6 lists the policies.

`Producer` and `ModificationDate` are output options: every merge writes them, and setting them pins the values that would otherwise describe the merge run. Neither can be set through `MetadataOverride`; `PdfMetadata.Producer` and `PdfMetadata.ModificationDate` are ignored for the result. Fixing the modification date removes one source of run-to-run difference. The output is reproducible only when the inputs, the modification date and the producer are the same; Chapter 6 explains the rest.

Split a large job into batches

Every source is held in memory until the merge ends, so a hundred large files can exhaust a process even though each one is below `MaxSourceBytes`. Merging in batches makes each intermediate job smaller and isolates a failure to one batch, but it does not make the final merge hold one batch: every batch file is a source of that merge too.

C#

```
static async Task MergeInBatchesAsync(
    IPdfMerger merger,
    IReadOnlyList<string> files,
    string destination,
    int batchSize,
    CancellationToken cancellationToken)
{
    if (files.Count == 0)
    {
        throw new ArgumentException("There must be at least one file to merge.", nameof(files));
    }

    if (batchSize < 1)
    {
        throw new ArgumentOutOfRangeException(nameof(batchSize), "The batch size must be at least one.");
    }

    var workDirectory = Path.Combine(Path.GetTempPath(), $"sharpcorepdf-{Guid.NewGuid():N}");
    Directory.CreateDirectory(workDirectory);

    try
    {
        var batches = new List<string>();

        for (var offset = 0; offset < files.Count; offset += batchSize)
        {
            var plan = MergePlan.Create();

            foreach (var file in files.Skip(offset).Take(batchSize))
            {
                plan = plan.AddFile(file);
            }

            var batchPath = Path.Combine(workDirectory, $"batch-{batches.Count:D4}.pdf");
            await merger.MergeToFileAsync(plan, batchPath, cancellationToken);
            batches.Add(batchPath);
        }

        var final = MergePlan.Create();

        foreach (var batch in batches)
        {
            final = final.AddFile(batch);
        }

        await merger.MergeToFileAsync(final, destination, cancellationToken);
    }
```

```
    finally
    {
        if (Directory.Exists(workDirectory))
        {
            Directory.Delete(workDirectory, recursive: true);
        }
    }
}
```

The cost is reading and writing the bytes twice and losing some document-level features twice: each intermediate merge applies the default policies, so page labels are already gone before the final merge. The final merge still reads and holds every batch, so peak memory is the total size of the batch files, not the size of one batch. If that total is too large, produce several smaller output documents, select fewer pages per document, or give the process more memory; the engine cannot write one merged document without holding all of its sources. If document-level features matter, keep the batch count low enough that the final merge is acceptable.

Accept a document only after checking it

A preflight gate turns a class of runtime failures into a clear rejection:

C#

```
const int MaximumPages = 2_000;

var content = new PdfFileSource(candidatePath);
PdfDocumentInfo info;

try
{
    info = await inspector.InspectAsync(content, cancellationToken: cancellationToken);
}
catch (PdfSourceFormatException exception)
{
    return Reject($"not a readable PDF: {exception.Detail}");
}

if (!info.IsReadable)
{
    return Reject(info.RequiresPassword
        ? "the document needs a password"
        : "the document uses an unsupported security handler");
}

if (info.PageCount == 0)
{
    return Reject("the document has no pages");
}

if (info.PageCount > MaximumPages)
{
    return Reject($"the document has {info.PageCount} pages; the limit is {MaximumPages}");
}

if (info.Warnings.Any(warning => warning.Kind is
        PdfWarningKind.PageTreeMissing or PdfWarningKind.PageTreeCycle))
{
    return Reject("the page tree is damaged; fix the source before uploading it");
}

return Accept(info);
```

A few warnings deserve rejection, most do not. `PageTreeMissing` means the document contributes no pages; `PageTreeCycle` means a page tree pointed back at itself. A repaired cross-reference table, an assumed page size or a truncated object stream on an otherwise readable document is usually a reason to log, not to refuse the file. Chapter 8 lists every kind with its meaning.

Notice that `PdfSourceFormatException` is caught even though inspection is supposed to report rather than throw. It still throws when the bytes are not a PDF at all or the limits are exceeded — those are not "encrypted but closed" conditions that a `PdfDocumentInfo` can describe.